

LESSON 5-3 • SESSION 1 • PRACTICE • FORM B

Trapezoids by Decomposing

Learning target – I can find the area of a trapezoid by decomposing it into a rectangle and triangles. · 6.G.A.1

REMEMBER

- A **trapezoid** has one pair of parallel sides, called the **bases**. The **height** meets the bases at a square corner.
- **Decompose**: cut it into a rectangle and one or two triangles. Find each area, then **add**.
- The triangle bases together equal the **long base – short base**. Two matching triangles split it in half.

WORKED EXAMPLE

Bases 6 cm and 14 cm, height 5 cm, two matching triangles.

Step 1 Rectangle: $6 \times 5 = 30 \text{ cm}^2$.**Step 2** Each triangle base: $(14 - 6) \div 2 = 4 \text{ cm}$.**Step 3** Triangles: $2 \times (\frac{1}{2} \times 4 \times 5) = 20 \text{ cm}^2$.**Answer** $30 + 20 = 50 \text{ cm}^2$ **WORDS** trapezoid (*trapezio*) · decompose (*descomponer*) · height (*altura*)

PRACTICE

show your work · label units

1. Each trapezoid is cut into a rectangle and two matching triangles. Find the base of one triangle.

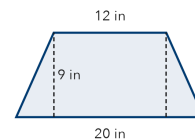
a. bases 12 in and 20 in _____

b. bases 6 m and 16 m _____

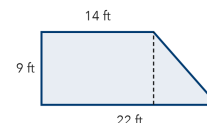
c. bases 9 cm and 21 cm _____

d. bases 15 ft and 23 ft _____

- 2.
- Craft Tabletop.**
- A craft tabletop is an isosceles trapezoid. It is cut into a rectangle and two matching triangles.

a. Area of the rectangle _____ in²b. Area of both triangles _____ in²c. Area of the tabletop _____ in²

- 3.
- Right-Angle Deck.**
- A deck is a trapezoid with one square corner. Decompose it into a rectangle and a triangle. Find its area.

Hint One triangle only. Its base is the long side minus the short side.

Show your work.

Answer _____ ft²

- 4.
- Bus Shelter Roof.**
- A roof panel is an isosceles trapezoid with bases 6 m and 16 m and a height of 8 m. Find its area.

Hint Split the leftover $16 - 6$ into two equal triangle bases.

Show your work.

Answer _____ m²

- 5.
- Corner Garden.**
- A garden is a trapezoid with one square corner. Its parallel sides are 9 yd and 15 yd, and the height is 14 yd. Find its area.

Show your work.

Answer _____ yd²

- 6. Tray and Mat.** A lunch tray is an isosceles trapezoid with bases 14 in and 24 in and a height of 12 in. A placemat is a rectangle 19 in by 12 in. Find each area. Do they cover the same space?

Hint Find the tray by decomposing, then the mat as length $\times$ width.

Show your work.

Same or not _____

- 7. Stage Apron.** A stage apron is an isosceles trapezoid with bases 15 m and 25 m and a height of 8 m. Decompose it and find its total area.

Hint Slice off the two end triangles. A rectangle is left in the middle.

Show your work.

Answer _____ m²

- 8. Deck Plan.** A deck is a trapezoid with bases 12 ft and 20 ft and a height of 5 ft. Decking costs \$9 per square foot.

- a. Area of each piece _____
b. Total area _____ ft²
c. Total cost _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Amina's Work.** A trapezoid has bases 13 m and 23 m and a height of 8 m. Amina cut it into a rectangle and two triangles, then wrote: "Area = $13 \times 8 = 104$ m²." Explain the mistake and find the correct area.

TEST PRACTICE

choose the best answer

- 10.** Every piece of a decomposed trapezoid has the same –
A base **B** area **C** height **D** slanted side
- 11.** A rectangle 8 cm by 5 cm plus a triangle with area 12 cm² gives a total area of –
A 25 cm² **B** 40 cm² **C** 52 cm² **D** 64 cm²

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

228 I	168 T	162 E	144 S	128 X	720 N	160 O	120 Z	88 C	200 Q
#2c	#3	#4	#5	#6	#7	#8c			

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____